

# Suanne Sparrow

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## Engineering Project Portfolio

Bachelor of Applied Science, Manufacturing Engineering , University of British Columbia, Class of 2026

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### Wind Turbine SHM System

Low-Cost Vibration Monitoring & IoT Networking

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C · ESP32 · FreeRTOS · Ethernet · MQTT

### Digital Twin

DeltaV Simulation, HMI & Control Narrative

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DeltaV Distributed Control Systems · HMI · Digital Twin

### MacBook Pro Chassis

CAM Process Design & Manufacturing

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Siemens NX · 5-Axis CNC · DFM

### Precision Weight Scale

Op-Amp Design & MATLAB DAQ

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Op-Amp Design · MATLAB · Arduino

## Monitoring System (Capstone 1<sup>st</sup> Place)

<https://github.com/pulkitm21/SHM-Capstone.git>

## Low-cost vibration monitoring & IoT networking

## OBJECTIVE

Design and validate a low-cost structural health monitoring prototype for wind turbines in seismic-prone regions, integrating embedded MEMS sensing, wired network transmission, local data storage, and secure remote dashboard access for continuous vibration monitoring.

## MY CONTRIBUTION: FIRMWARE DEVELOPMENT

**Sensor drivers:** implemented SPI drivers for ADXL355Z (accel.) & SCL3300 (incl.) and I<sup>2</sup>C driver for ADT7420 (temp.) in ESP-IDF from register-level documentation

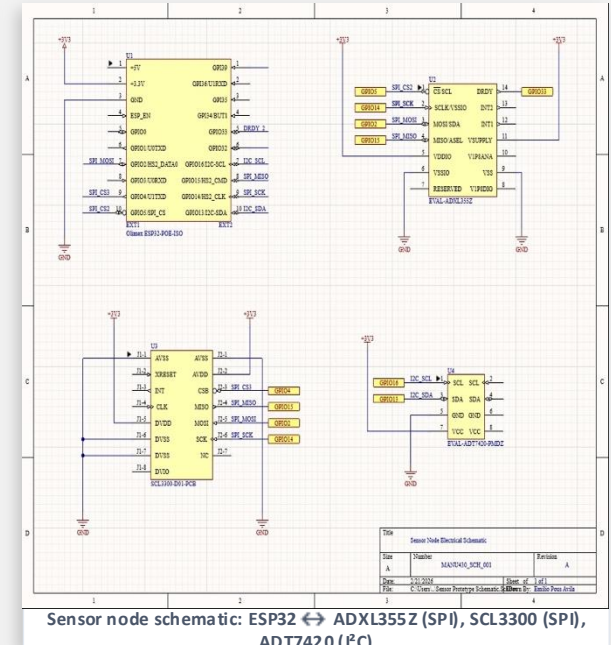
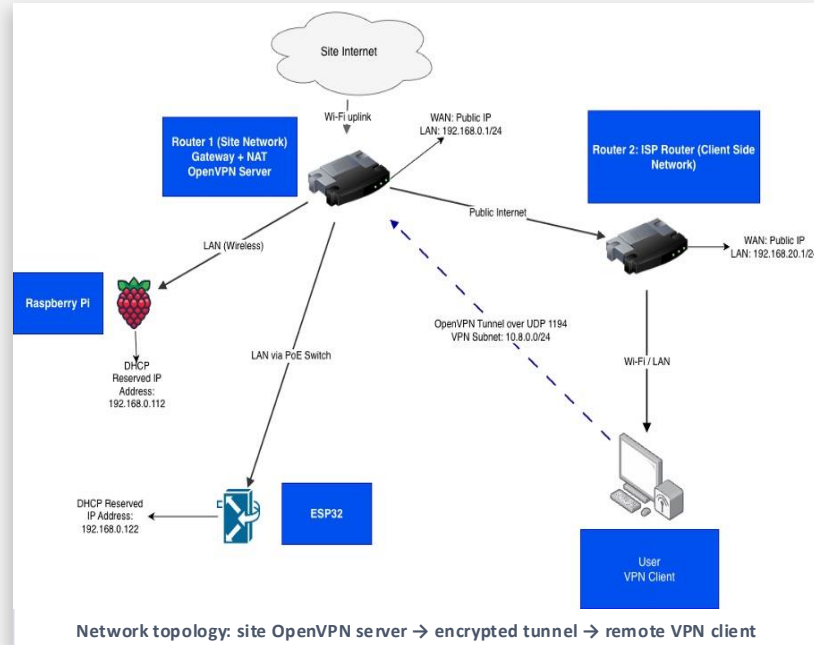
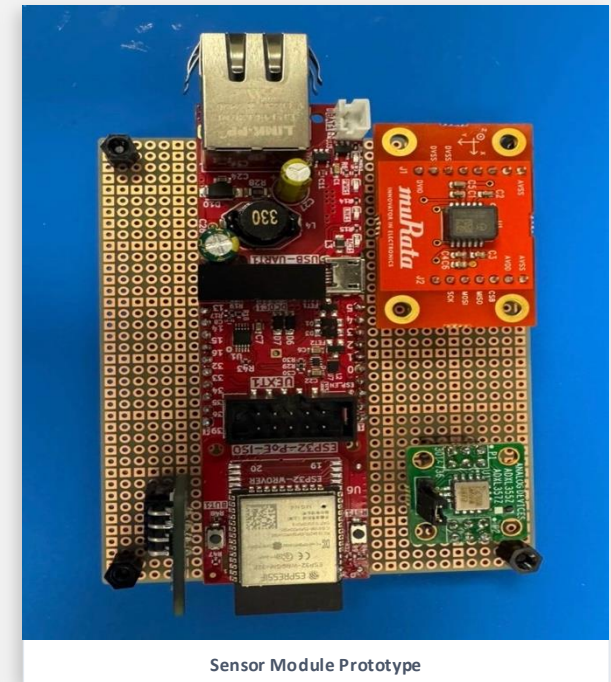
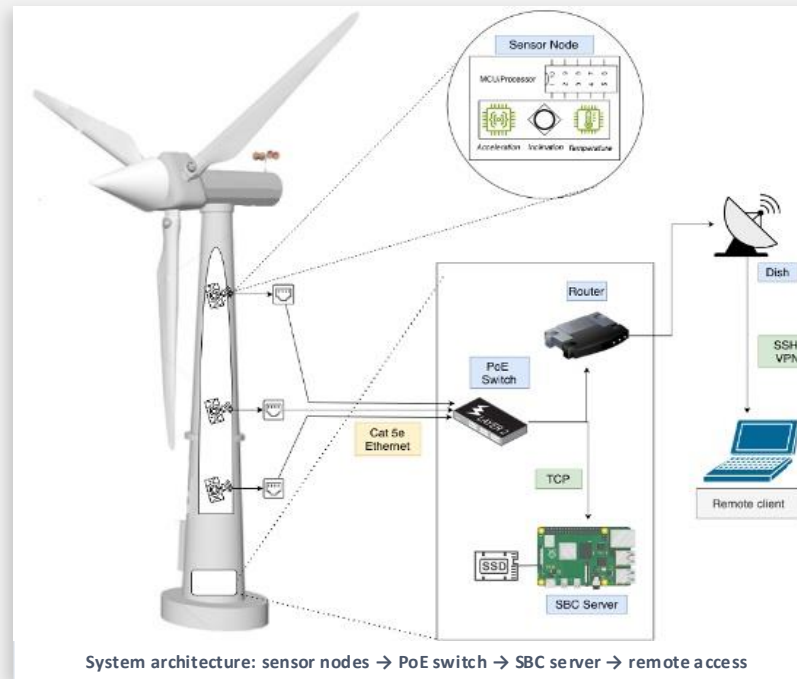
**Real-time acquisition pipeline:** designed timer/ISR-based sampling with lock-free ring buffers to separate deterministic sensor acquisition from FreeRTOS processing and network transmission

**Ethernet & PoE bring-up:** configured ESP32 Ethernet stack over PoE using LwIP; handled MAC/PHY initialization and network interface management in ESP-IDF

**MQTT + fault handling:** implemented MQTT client for structured sensor data, status, and fault publishing; added reconnect logic, dropped-sample logging, and NaN handling for partial sensor failures

## OUTCOME

Delivered a functional end-to-end prototype from sensor acquisition to Raspberry Pi storage and remote dashboard visualization. Validated simultaneous 7-channel acquisition, fault detection and propagation to the UI, and 24-hour operation with 99.83% data integrity.



# District Energy Digital Twin

DeltaV Simulation · HMI Development · Control Narrative

## OBJECTIVE

Build a control system and digital twin of the Senákw District Energy System (DES) in Emerson DeltaV Distributed Control System (DCS) - a 32 MW heating / 15 MW cooling plant serving Canada’s largest Indigenous development, enabling control verification, PID tuning, and operator training before physical commissioning.

## MY CONTRIBUTION

**Simulation modules:** built first-principles DeltaV composite blocks for heat exchangers, pumps, valves, and energy transfer stations (ETS), etc. simulating flow, pressure, and temperature behaviour

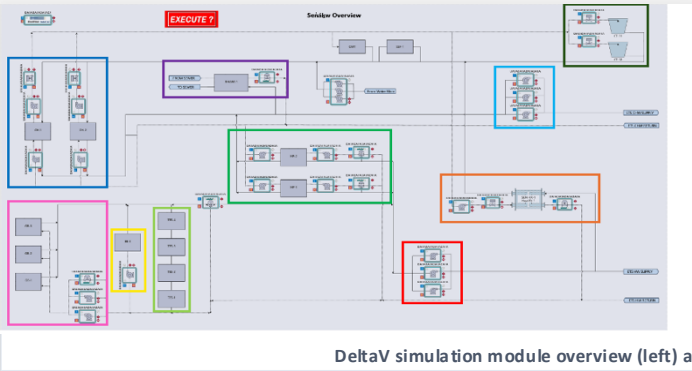
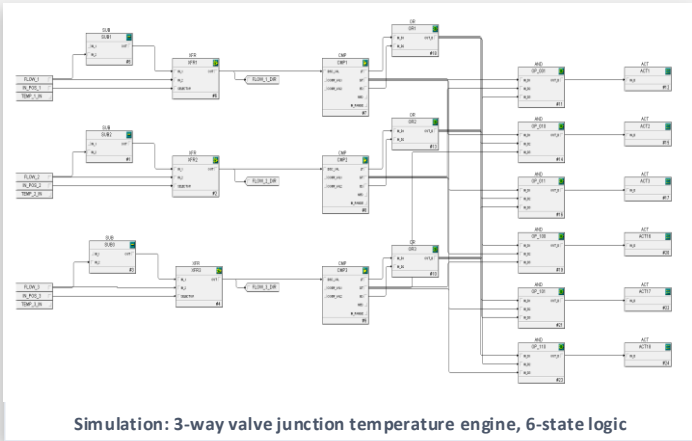
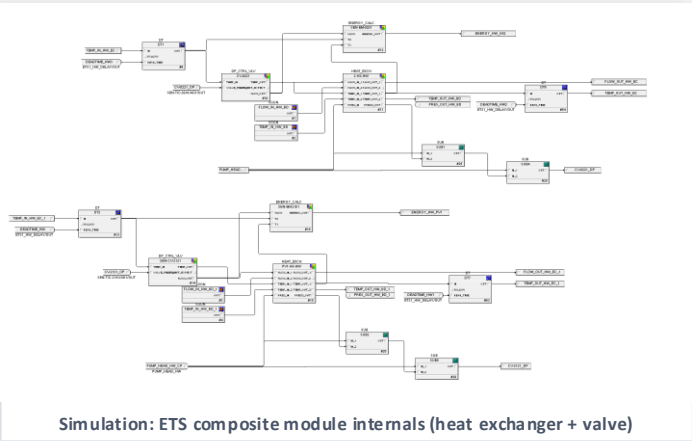
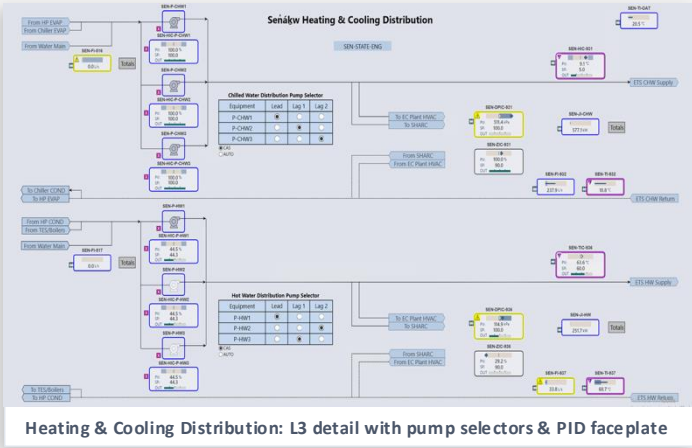
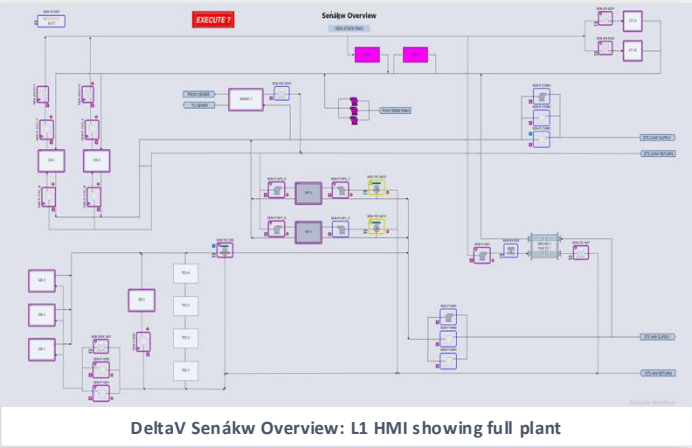
**HMI ownership (DeltaV Live):** designed all operator graphics to ISA-101 standard (L1–L4 hierarchy); custom GEMs for each equipment class

**Control narrative reference:** studied the control narrative to verify the control logic was correctly implemented, and to identify any discrepancies requiring design changes

**Control design validation:** identified and resolved HP circulation pump issue via simulation; variable speed fix implemented to eliminate chiller recirculation inefficiency

## OUTCOME

Digital twin operational for all 3 buildings and full energy center. Simulation revealed control design flaw corrected pre-commissioning. HMI delivered covering L1–L4 with ISA 101-compliant grey graphics.



# MacBook Pro Keyboard Chassis

CAM Process Design & Manufacturing Engineering

## OBJECTIVE

Design the complete CNC manufacturing process for a MacBook Pro keyboard aluminum chassis using Siemens NX. From fixturing strategy through CAM programming, producing a verified In-Process Workpiece across two machining setups (top & bottom).

## MY CONTRIBUTION

**Fixture design:** vacuum workholding (bottom) + mechanical clamping (top); full 6-DOF analysis

**Tool & holder selection:** 7 Kennametal tools; custom T-cutter designed for undercut geometry

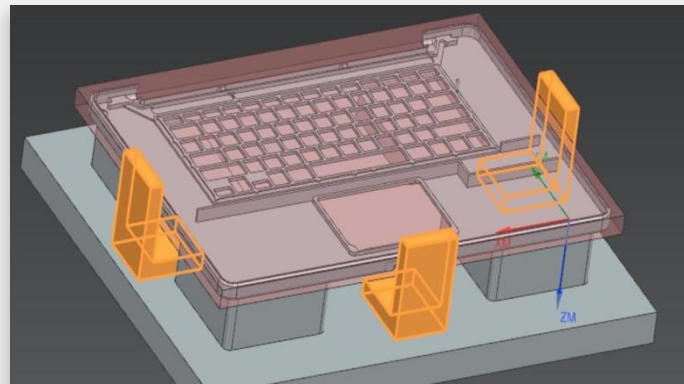
**Cutting parameters:** RPM, feed, MRR, power & torque calculated and verified within machine limits

**CAM programming:** 30 operations in Siemens NX incl. 5-axis Variable Streamline contouring

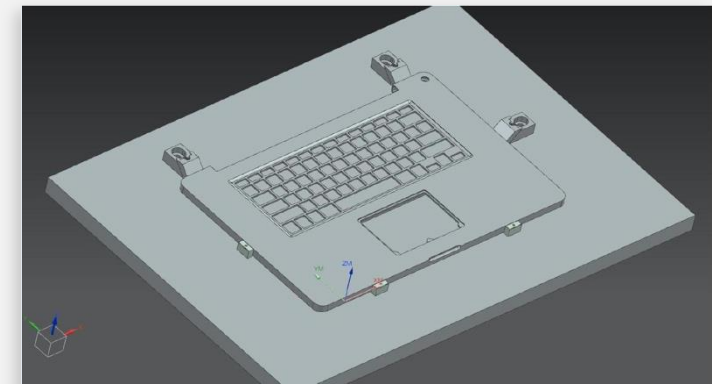
**DFM analysis:** identified 5 unmanufacturable feature classes; issued redesign recommendations

## OUTCOME

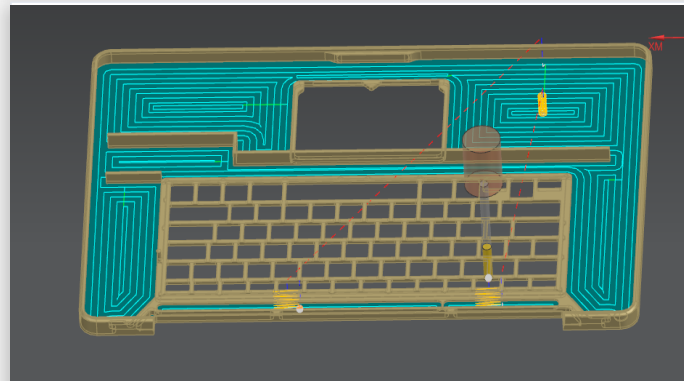
IPW verified. Zero gouges, no remaining stock, achieved  $\pm 0.05$  mm tolerance. 30 CAM operations across 2 setups. DFM feedback provided on sharp corners, tight clearances, and undercut access.



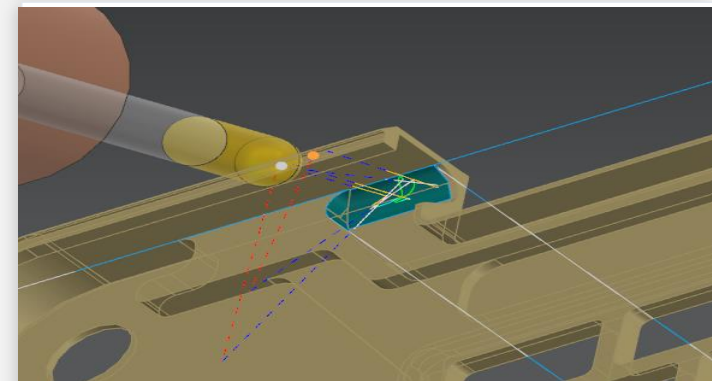
Full bottom vacuum fixture setup with removable side stops



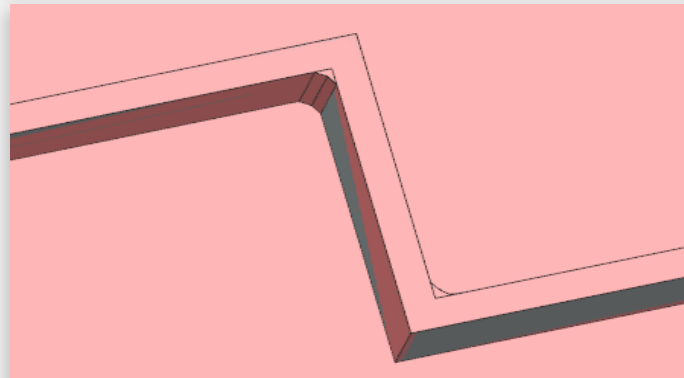
Top fixture setup with 3 low profile side clamps and 3 locating keys



Follow Periphery strategy for floor and wall



Variable streamline to cut angled grooves with 3D curvature



Unmanufacturable feature



Final IPW: bottom side (all operations complete)

# Precision Strain Gauge Weight Scale

Op-Amp Circuit Design, Simulink Verification & MATLAB DAQ

## OBJECTIVE

Design, build, and program a complete strain-gauge weight scale from raw millivolt sensor signal through analog conditioning to a calibrated real-time mass readout in MATLAB.

## MY CONTRIBUTION

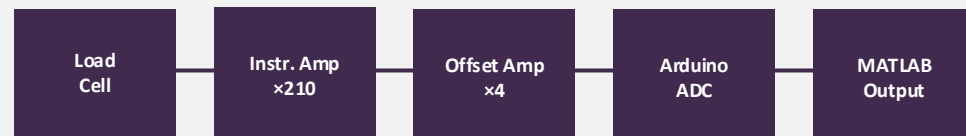
**Instrumentation amplifier:** full  $V_{out}$  derived via nodal analysis

**Simulink verification:** circuit modelled to confirm gain before physical build;  $V_{out} = 1.45\text{ V}$  and gain = 210 confirmed

**Offset amplifier:** solved  $2 \times 2$  resistor system to scale output to Arduino ADC range (0.5–2.5 V)

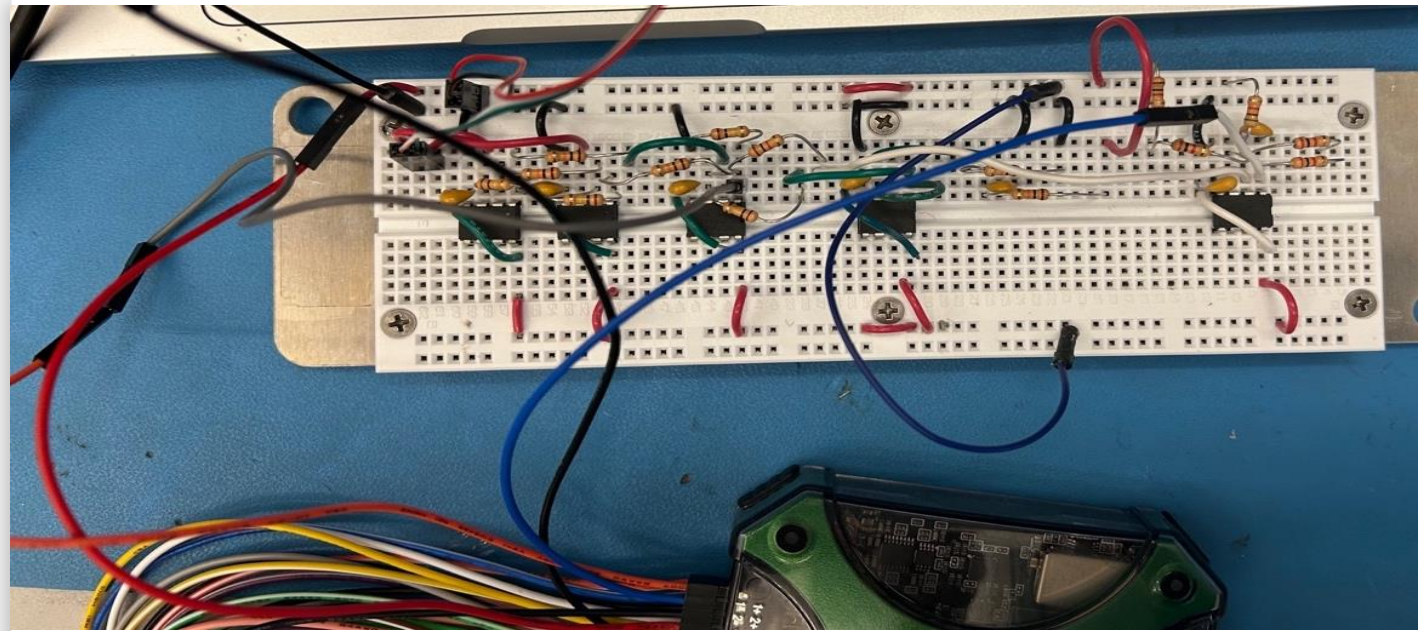
**MATLAB DAQ program:** 100 Hz acquisition loop; calibrated equation mass =  $1.11\text{ V} - 0.591$  validated at 0 & 2 kg

## SIGNAL CHAIN

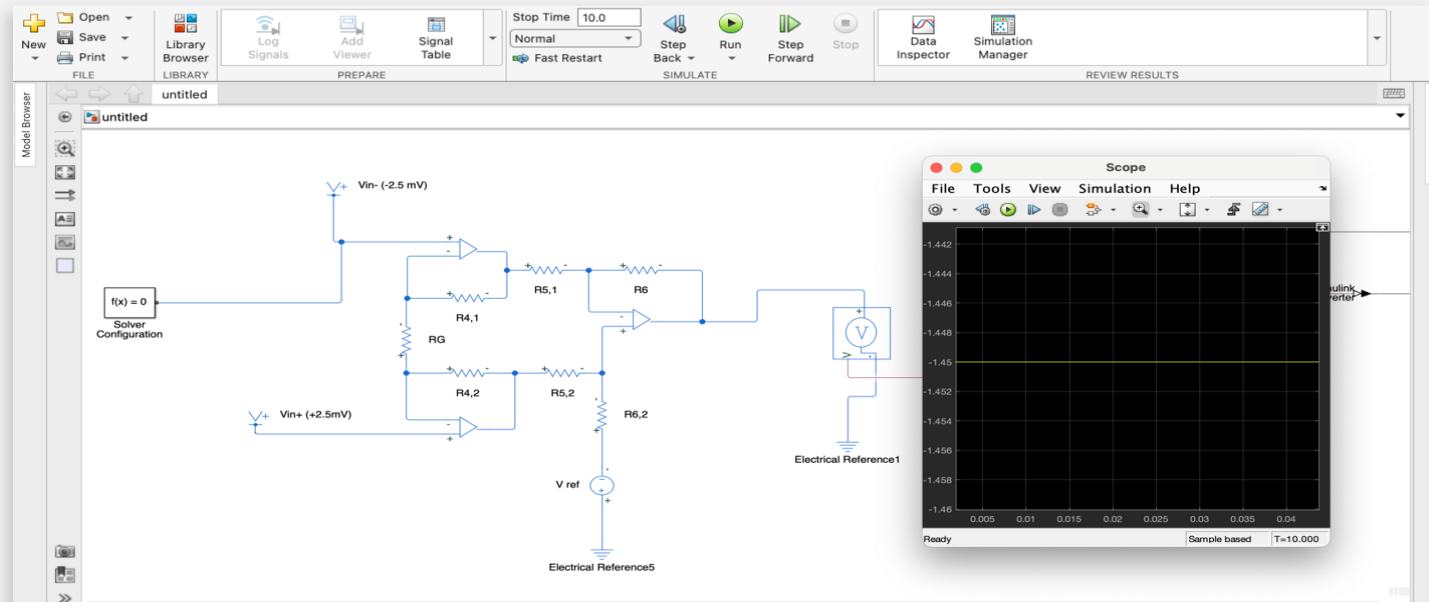


## OUTCOME

Measured gain = 210. Output 0.62–2.41 V (target 0.5–2.5 V). Mass readout validated at 0 kg & 2 kg. Minor deviation from standard resistor value substitutions.



Assembled instrumentation + offset amplifier circuit on breadboard with function generator



Simulink model verifying gain = 210 before physical breadboard build